# Computing the Astrometric Coordinates of Bright Asteroids over the Epoch of the DASCH Photometric Plates

Presentation by Ignacio M. Pineda.

Project done in collaboration with Dr Zach Murray, Dr Matthew Holman and Peter Williams.

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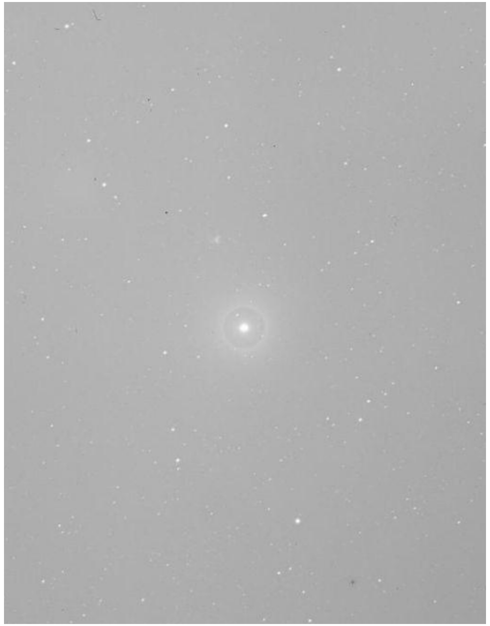
Omega Centauri on plate a01735



Betelgeuse as can be seen on plate a00367

# 1. DASCH Project

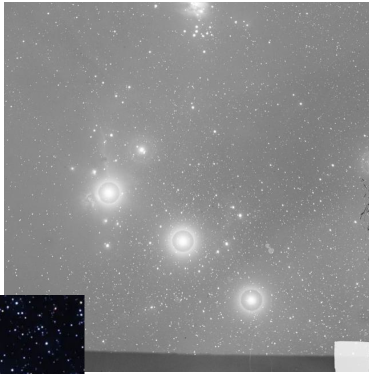
- Plates located at the Harvard-Smithsonian CFA

- Photographic plate: Sheet of glass covered with chemical susceptible to light. Creates image

- Digitizing the plates immortalizes them and makes them accessible to everyone

- Nearly a hundred years of work (1880-1990)

- Offers the <u>first complete image of the observable universe</u>



Orion constellation, on plate b2312

Halley's comet on plate b41215

# Important figures involved in the Dasch project:

Cecilia Payne-Gaposchkin
Discovered chemical composition of stars

Henrietta Leavitt
Measured distance in space

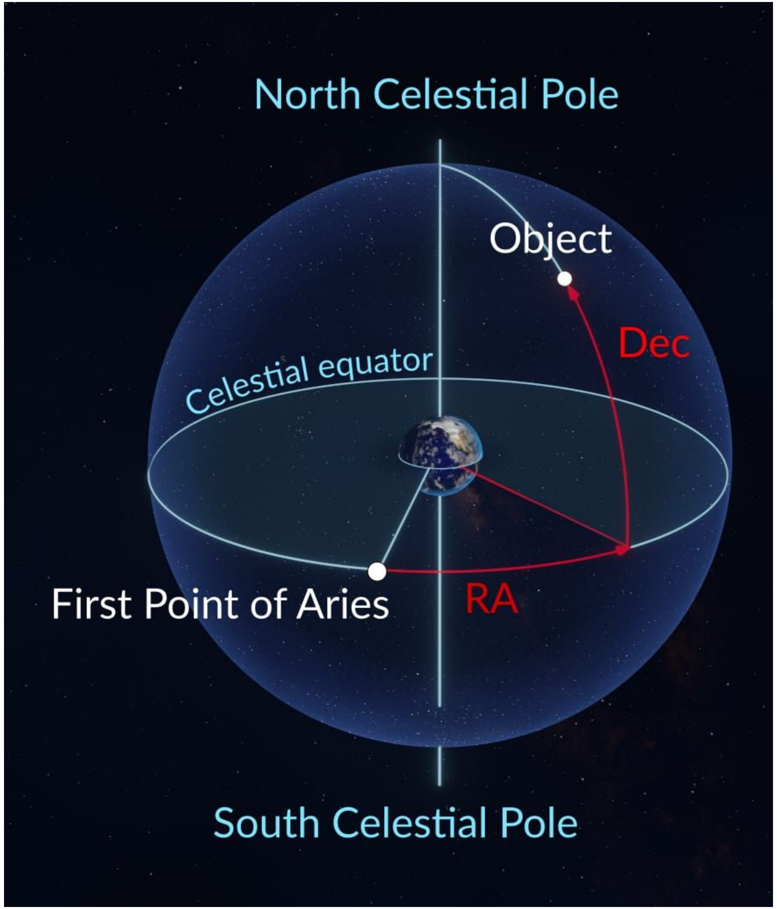
Annie Jump Cannon
Classification of stars based on temperature

# 2. Project Goal

**Computing** the astrometric coordinates of asteroids with a semi-major axis of 2.5 AU

We require the ra/dec coordinates

We must compute the astrometric coordinates via computational integration




Source: Star Walk

# 3) Process of the project

## 3.1) Integrating the orbits and obtaining the astrometric coordinates

- Takes the orbit of an asteroid as a function of distance:

$$m_i \frac{d^2\mathbf{r}_i}{dt^2} = G \sum_{j=1 j\neq i}^{n} m_i m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|^3}, \quad i = 1, 2, \ldots, n$$

- Solves the N-body problem via integration with Assist
  - Assist relies on IAS-15, a 15th order integrator.
- Assumed that mass of asteroids is zero and there is no Yarkovsky effect
- To plot the predicted coordinates → two pass filtering algorithm devised by Dr Murray
  - Max ra/dec bounds of the image and kick anything outside those bound
  - If not rejected do a closer path by projecting full bounds of the image


Source: The MPC

# 3.2) The light-time correction

The light-time correction:

- Since time doesn’t travel instantly, there must be a correction rearranged:
    - te = emission time (unknown)
    - tr = reception or observation time
    - r(t) = Displacement vector at that time

$$t_r = t_e + \frac{|\mathbf{r}(t_e) - \mathbf{r}_{\text{obs}}(t_r)|}{c}$$

$$\boxed{t_e = t_r - \frac{|\mathbf{r}(t_e) - \mathbf{r}_{\text{obs}}(t_r)|}{c}}$$

- It is an implicit equation and it is impossible to solve for tr. We use recursion until the difference is negligible.

- Converges quickly because most velocities satisfy:

$$\frac{v}{c} << 1$$

- Accurately obtain the object's position

$$t_e^{(0)} = t_r - \frac{|\mathbf{r}(t_r) - \mathbf{r}_{\text{obs}}(t_r)|}{c}$$

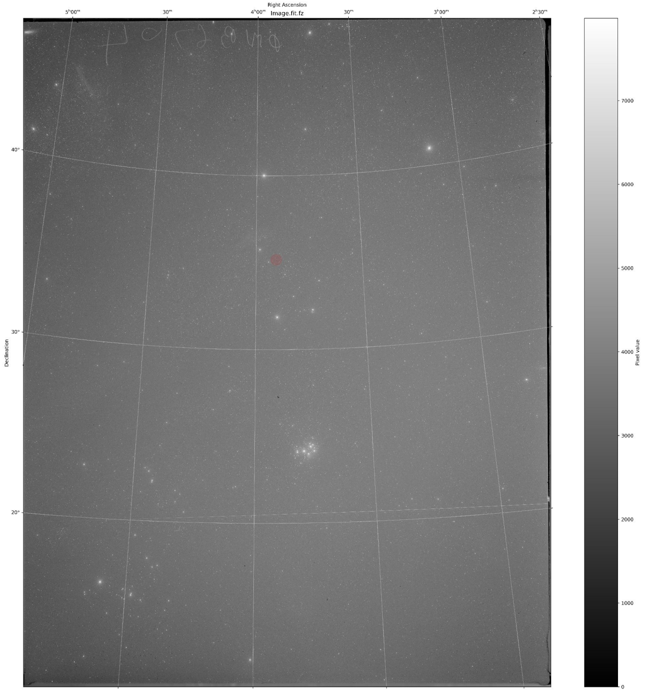
$$\boxed{t_e^{(n+1)} = t_r - \frac{\left|\mathbf{r}(t_e^{(n)}) - \mathbf{r}_{\text{obs}}(t_r)\right|}{c}}$$

# 3.3) Plotting the coordinates

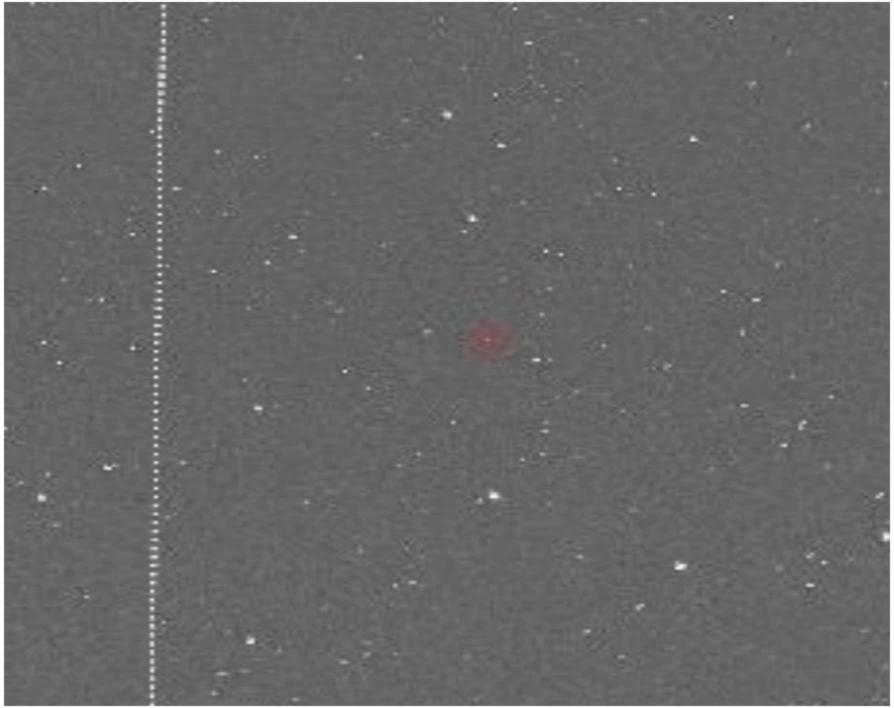
- I computed ≈6500 asteroids over a 100 year time-span, producing ≈60 GB of data
- We obtain a json file with the asteroids that have a time span and coordinates that coindice on a plate, with the following information:
    - Julian date of the plate - 1,1, 4713 BC
    - The asteroid plate
    - ra/dec coordinates

# 3.4) Plotting the coordinates

- Built a sample code plotted on several plates. <u>All were successful</u>
- All results on my GitHub
- Asteroids have more apparent motion → They can appear as streaks



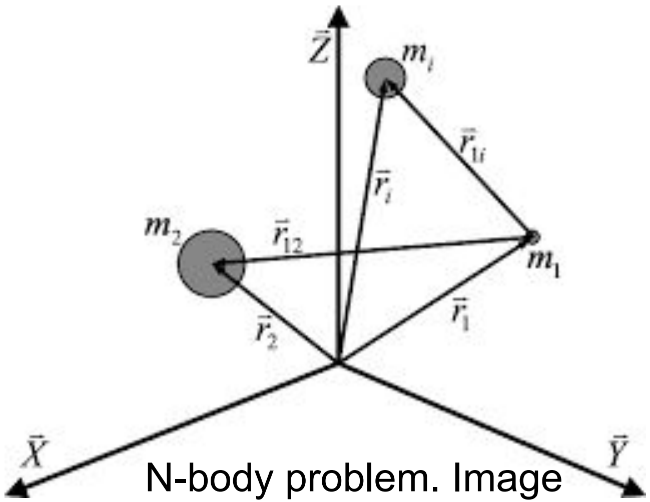

Example of a valid result with plate **dnb06704**



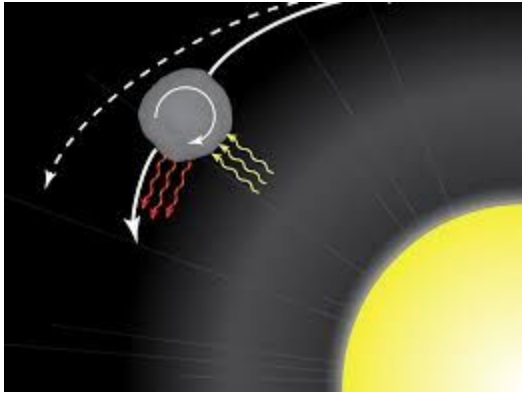
Example of a valid result with plate **dnb06678**

# 3.5) Limits

- Limits presented by the Dasch project: Only apparent magnitude under 17 is detectable via Dasch plates
- Space for error due to following approximations:

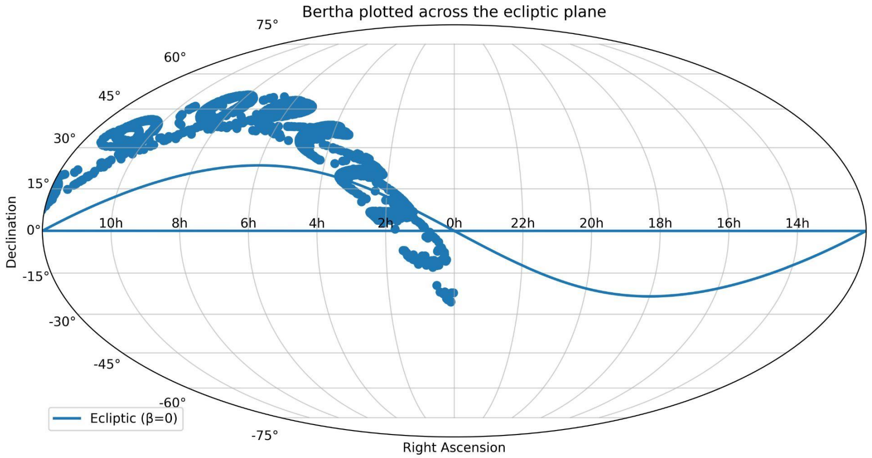


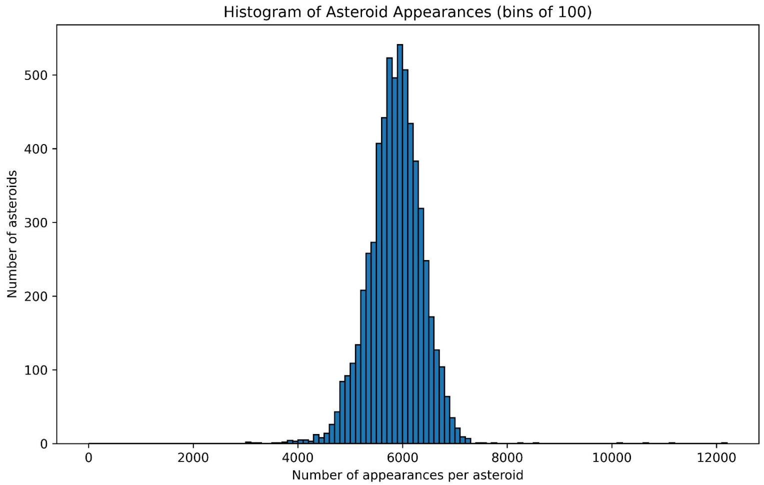

N-body problem. Image taken from ResearchGate

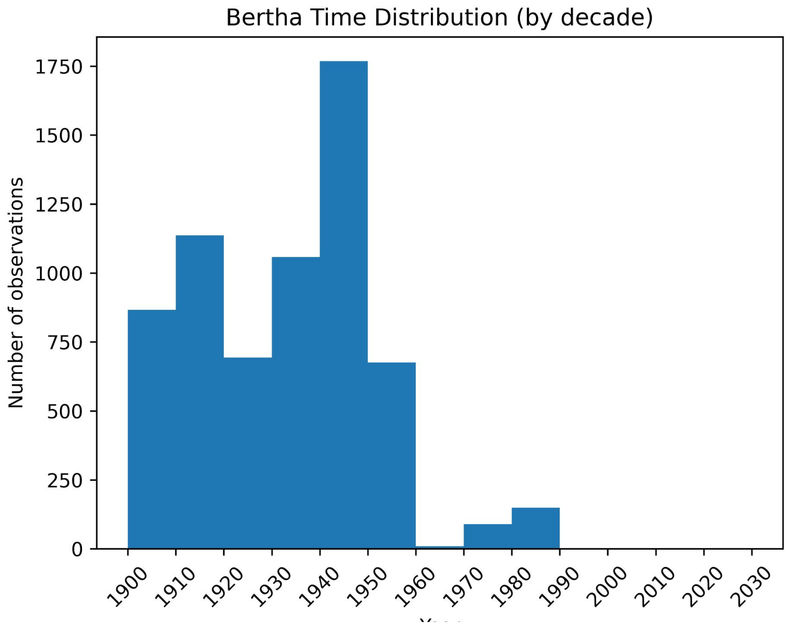


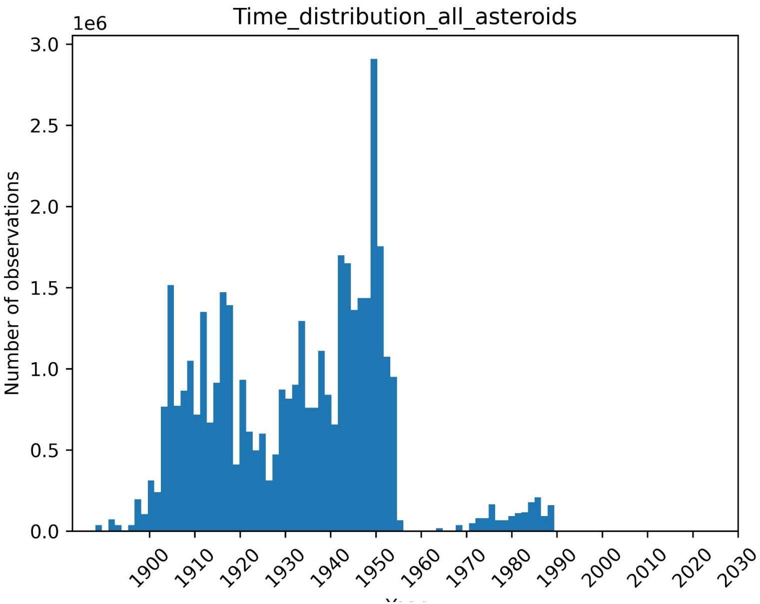
Yarkovsky effect. Property of Dr. Lauretta, University of Arizona

# 4) Statistics



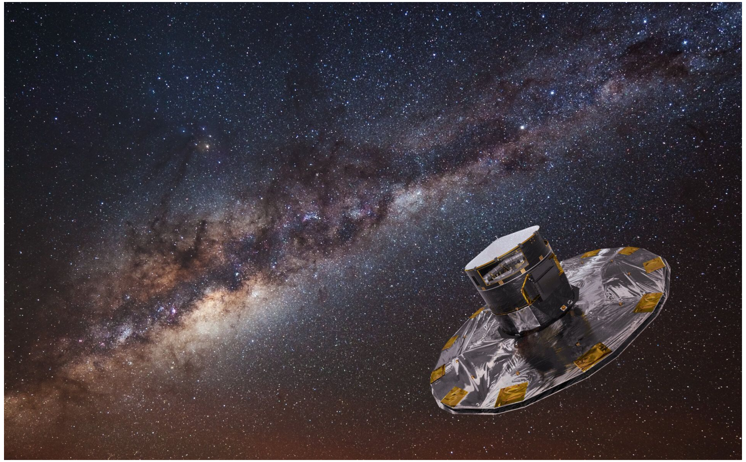

Bertha Time Distribution (by decade)
Number of observations
0
250
500
750
1000
1250
1500
1750
1900
1910
1920
1930
1940
1950
1960
1970
1980
1990
2000
2010
2020
2030


Time_distribution_all_asteroids
1e6
Number of observations
0.0
0.5
1.0
1.5
2.0
2.5
3.0
1900
1910
1920
1930
1940
1950
1960
1970
1980
1990
2000
2010
2020
2030

# All of the mentioned code is on GitHub

# Opportunities

If an asteroid is not found where we predict: one of our assumptions must be wrong



An illustration of Gaia. ESA

Similar study conducted at JPL by Dr Oscar Fuentes-Muñoz at JPL:

- JPL - Jet Propulsion Laboratory
- Gaia - One of ESA's surveyors
- Gives more exact measurements by using Gaia + MPC
- Only uses modern data (2015-2020 review data)

# 5. Future impact

- Allows for potential astronomical precovery Finding an object in an image dated before the object's official discovery

- Facilitates the obtainment of the real coordinates, coming from the predicted coordinates.

- Offers new data on which to base future projects

- Contributions to the study of photometry and astrometry in the solar system (include explanation in script, further up)

Linkedin: Ignacio M. Pineda

GitHub: IgnacioMPineda

https://github.com/IgnacioMPineda/Astrometric-Coordinates-of-Bright-Asteroids-over-the-Epoch-of-the-DASCH-Photometric-Plates

# Q&A